SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Technical Presentation

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Technical Presentation

Question 1 (5 Marks)
Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing

Question 2 (5 Marks)
Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel

Question 3 (5 Marks)
Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing

Question 4 (5 Marks)
A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination

Question 5 (5 Marks)
Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total MarkS
Q1						
Q2						
Q3						
Q4						
Q5						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________